Research Project - IT4010 Data Analysis Report

ID Number: IT22083982

Project Title: Nutritional Guidance – Predictive Health Intervention System

Introduction

1.1. Background:

Modern nutrition tracking systems primarily focus on short-term calorie monitoring or daily nutrient summaries, often overlooking long-term dietary behavior. However, micronutrient deficiencies such as iron, calcium, protein, and energy imbalance develop gradually over extended periods and frequently remain undetected until visible health symptoms appear. In Sri Lanka, preventive nutritional screening and personalized dietary analytics remain limited, leaving individuals without early warning mechanisms for hidden nutrient imbalances. Most existing applications fail to monitor longitudinal eating patterns, compare intake against age-specific nutritional requirements, predict future deficiency risks, or provide meaningful personalized health intervention insights. Micronutrient deficiencies such as iron and calcium deficiency remain significant public health concerns in developing countries, particularly among vulnerable age groups. Early predictive identification using dietary analytics can reduce long-term health complications.

The Nutritional Guidance – Predictive Health Intervention System addresses this gap by leveraging data analytics and machine learning techniques to analyze longitudinal dietary intake data. The system aggregates user eating patterns over time, evaluates nutrient adequacy against recommended standards, and classifies deficiency risk levels (Low, Medium, High) before clinical symptoms emerge. By shifting from reactive nutrition tracking to proactive predictive health monitoring, this system enables early detection of potential deficiencies and supports personalized, data-driven dietary intervention strategies.

1.2 Component Scope within the AI-Powered Kitchen Ecosystem:

This Nutritional Guidance component operates as the health intelligence layer of the AI-Powered Kitchen Ecosystem. It consumes food intake logs generated through user tracking and food database integration, then produces nutrient adequacy analytics and deficiency risk classifications. These outputs support personalized dietary recommendations and early preventive intervention, complementing other ecosystem modules such as food expiry prediction and smart shopping assistance.

2.1. Research Problem:

Existing nutrition tracking systems primarily analyze daily intake in isolation without considering long-term dietary behavior. They do not evaluate temporal eating patterns across weeks or months, making it difficult to detect gradual nutrient imbalances. Furthermore, most systems fail to predict future nutritional risks and instead provide generalized recommendations that lack personalization based on age, health conditions, or historical intake trends.

As a result, users remain unaware of developing micronutrient deficiencies such as iron, calcium, protein, or energy imbalance until adverse health symptoms appear. The fundamental research problem addressed in this study is the absence of a data-driven, predictive, and personalized mechanism capable of forecasting nutrient deficiency risks using longitudinal eating pattern analysis.

2.2. Objectives:

Objective	Objective number
Analyze user dietary intake patterns over time	1

Identify long-term nutrient imbalance trends	2
Predict potential nutrient deficiency risks using machine learning	3
Generate personalized nutritional risk insights using adjustment factors	4

2.3 Research Questions

RQ1: How accurately can longitudinal dietary intake patterns predict nutrient deficiency risk levels (Low/Medium/High) using machine learning?

RQ2: To what extent do nutrient adequacy ratios (intake ÷ requirement) improve risk classification compared to using raw nutrient intake values alone?

RQ3: How effectively can age-based requirements and health-condition adjustment rules support personalized risk estimation in a preventive nutrition monitoring system?

3. Data Exploration

3.1. Data Collection

Two primary data collection approaches were used:

- **Structured Nutrition Datasets**

  Publicly available and curated nutrition datasets were obtained from reliable sources, including Kaggle-hosted food nutrition datasets, providing nutrient composition information for Sri Lankan and regional food items.
- **User Dietary Intake Records**

  Simulated and system-generated food consumption logs were used to represent user dietary intake, aggregated over weekly and monthly periods to support temporal eating pattern analysis.

3.2. Dataset Description:

Data Source	Description	Resource	Size	Key Attributes
Food Nutrition Dataset	Nutritional composition of food items	Kaggle Dataset	~1200 foods	energy, protein, calcium, iron
Nutrient Requirement Dataset	Age-based recommended daily nutrient intake values	World Health Organization (WHO) Guidelines & National Nutrition Standards	15 age groups	Recommended daily intake
Health Adjustment Dataset	Nutrient requirement adjustments for specific health conditions	Clinical Nutrition Guidelines	10+ conditions	adjustment factor

Intake History Dataset	User food consumption and intake logs	System-generated user interaction data	Time-series	food_id, quantity, date

Following preprocessing and synthetic intake generation procedures, a total of 3,200 labeled dietary samples were constructed for supervised learning. The generated dataset ensured balanced representation across LOW, MEDIUM, and HIGH deficiency risk categories.

3.3. Suitability Analysis

3.3.1. Relevance to Individual Research Objectives:

All datasets align with project objectives:

Objective	Nutrition Dataset	Requirement Dataset	Adjustment Dataset	Intake Logs
Analyze user intake patterns over time	✓	✓		✓
Identify long-term nutrient imbalance trends	✓	✓		✓
Predict potential nutrient deficiency risks	✓	✓	✓	✓
Generate personalized nutritional risk insights	✓	✓	✓	✓

4. Methodology

4.1. Data Preprocessing:

Transformation Technique	Nutrition Dataset	Requirement Dataset	Adjustment Dataset	Intake Logs
Data Cleaning	✓	✓	✓	✓
Handling Missing Data	✓	✓	✓	✓
Data Type Conversion	✓	✓	✓	✓
Data Normalization	✓	✓		✓
Feature Engineering	✓	✓	✓	✓
Data Encoding		✓	✓	
Data Aggregation				✓

Dimensionality Reduction (PCA)				✓

4.2. Scalability

User dietary intake data grows continuously as the system is used, enabling comprehensive longitudinal analysis across extended time periods. The aggregation pipelines are specifically designed to efficiently process large-scale time-series intake datasets, ensuring that daily records can be transformed into meaningful weekly and monthly nutritional trends without performance degradation. This continuous data accumulation strengthens the system's ability to detect gradual nutrient imbalances and evolving dietary behaviors.

In addition, the nutrition datasets are extensible, allowing new food items and nutrient attributes to be incorporated without modifying the core system logic. The predictive models can also be periodically retrained using newly accumulated user profiles, improving classification accuracy and enabling the system to adapt to changing dietary patterns. This design ensures long-term scalability, adaptability, and sustained predictive performance.

4.3. Feature extraction and preprocessing

- User dietary intake records were aggregated to compute average daily nutrient intake based on user eating patterns.
- Temporal aggregation was applied to derive weekly and monthly nutrient intake trends, enabling long-term dietary behavior analysis.
- Nutrient intake values were compared against age-based recommended requirements to calculate intake-to-requirement ratios.
- Machine learning models were applied to classify nutrient deficiency risk levels (Low, Medium, High) using historical dietary intake patterns.
- Personalized adjustment factors were incorporated to account for age-related and health condition–specific dietary variations.

These extracted features effectively represent user eating behavior over time and support early risk detection through predictive health intervention.

4.4. Machine Learning Model Specification

The deficiency risk prediction component was implemented using a Random Forest Classifier (RandomForestClassifier) due to its robustness against noisy dietary intake patterns and its ability to model non-linear relationships between nutrient adequacy and risk outcomes. The model was configured with 250 decision trees (n_estimators=250), a maximum tree depth of 10 (max_depth=10), balanced class weights (class_weight="balanced") to mitigate class imbalance, and a fixed random seed of 42 (random_state=42) to ensure reproducibility. The trained model was serialized using joblib and deployed in the backend to support real-time risk predictions. Model validation included both stratified train–test evaluation and 5-fold cross-validation to ensure robustness and reproducibility.

4.4.1 Model Selection Justification

The Random Forest classifier was selected due to its robustness, stability, and suitability for structured nutritional data. As an ensemble method, it reduces variance through aggregation of multiple decision trees, minimizing overfitting risk. This is particularly important when modeling synthetic yet behaviorally diverse dietary intake samples.

Deficiency risk classification involves nonlinear threshold effects derived from adequacy ratios (intake ÷ requirement). Random Forest effectively captures these nonlinear interactions between nutrient totals, adequacy ratios, age, and health-condition adjustments without requiring complex feature engineering.

Compared to neural networks, Random Forest offers greater interpretability through feature importance analysis, which is essential in health-related applications where model transparency is critical. Additionally, it performs efficiently on moderate-sized datasets (n = 3,200) and demonstrates stable performance under cross-validation, making it an appropriate and reliable choice for this predictive nutritional framework.

4.5 Feature Set Definition

Each training sample was represented using a feature vector that captures both absolute intake and requirement-normalized adequacy signals. The final feature set included: age, total daily energy/protein/calcium/iron intake totals (total_energy_kcal, total_protein_g, total_calcium_mg, total_iron_mg), nutrient adequacy ratios computed as intake divided by requirement (ratio_energy, ratio_protein, ratio_calcium, ratio_iron), and a binary indicator of whether a health condition adjustment was applied (has_condition). This design supports explainability, as adequacy ratios directly indicate whether intake meets recommended thresholds.

4.6 Risk Labeling Strategy

Due to the limited availability of clinically validated nutrient deficiency labels, deficiency risk categories were generated using an interpretable rule-based heuristic derived from nutrient adequacy ratios. The adequacy ratio was computed as the proportion of actual intake relative to the recommended requirement (intake ÷ requirement).

A nutrient was classified as inadequate when its adequacy ratio fell below 0.80, consistent with common nutritional assessment practices where intake below 80% of recommended levels is considered nutritionally insufficient in population-level evaluations.

Daily deficiency risk labels were assigned based on the cumulative number of inadequate nutrients observed:

- 0 inadequate nutrients → LOW risk
- 1–2 inadequate nutrients → MEDIUM risk
- 3 or more inadequate nutrients → HIGH risk

This labeling framework provides a transparent and interpretable mapping between nutrient adequacy and risk severity. It enables consistent supervised machine learning training while preserving clinical interpretability of how risk categories are formed. Although heuristic-based, this approach establishes a structured approximation of deficiency risk in the absence of laboratory-confirmed biomarker data.

4.6.1 Mathematical Representation of Risk Labeling

Let:

$$R_i = \frac{Intake_i}{Requirement_i}$$

Define:

$$Deficient_i = \begin{cases} 1 & \text{if } R_i < 0.80 \\ 0 & \text{otherwise} \end{cases}$$

Total deficiency score:

$$D = \sum Deficient_i$$

Risk label:

$$Risk = \begin{cases} Low & D = 0 \\ Medium & 1 \leq D \leq 2 \\ High & D \geq 3 \end{cases}$$

4.7 Model Training and Validation Protocol

Synthetic dietary intake samples were generated by randomly selecting 3–8 food items from the master nutrition dataset and assigning realistic portion sizes ranging between 50 g and 300 g. For each age-specific requirement band, multiple intake profiles were created to enhance model generalization across demographic variations. In total, 3,200 synthetic daily intake samples were generated and labeled according to the defined deficiency risk strategy.

The dataset was partitioned using an 80:20 stratified train–test split, resulting in 2,560 samples for training and 640 samples for testing. Stratification preserved proportional representation of LOW, MEDIUM, and HIGH risk categories across both subsets. Model performance was evaluated using overall accuracy together with class-wise precision, recall, and F1-score metrics, ensuring balanced assessment across all deficiency risk levels.

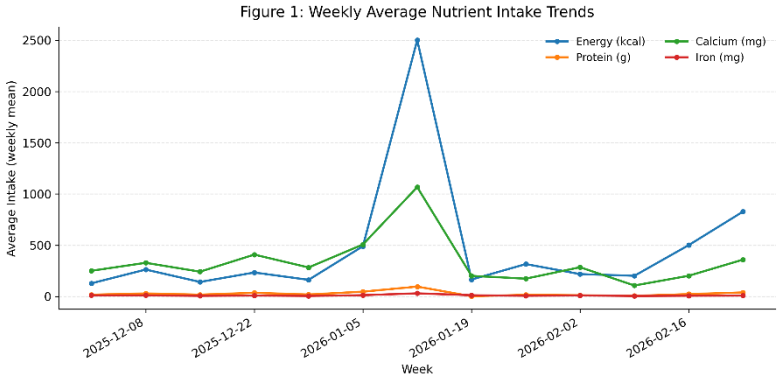
5. Modelling and Results

5.1. Key Insights:

a. Nutrient Intake Trend Analysis

Figure 1 illustrates the weekly average intake trends for energy, protein, calcium, and iron across the observation period. Unlike daily intake summaries, weekly aggregation reduces short-term fluctuations and highlights sustained dietary patterns. The results indicate noticeable variability in energy and protein intake, while micronutrients such as calcium and iron demonstrate more consistent but generally lower intake levels.

This temporal variation confirms that dietary behavior is not static and emphasizes the importance of longitudinal modelling. Short-term daily adequacy may mask cumulative deficiencies, whereas weekly patterns reveal persistent intake gaps. These findings support the need for time-series aggregation before applying deficiency risk prediction models.

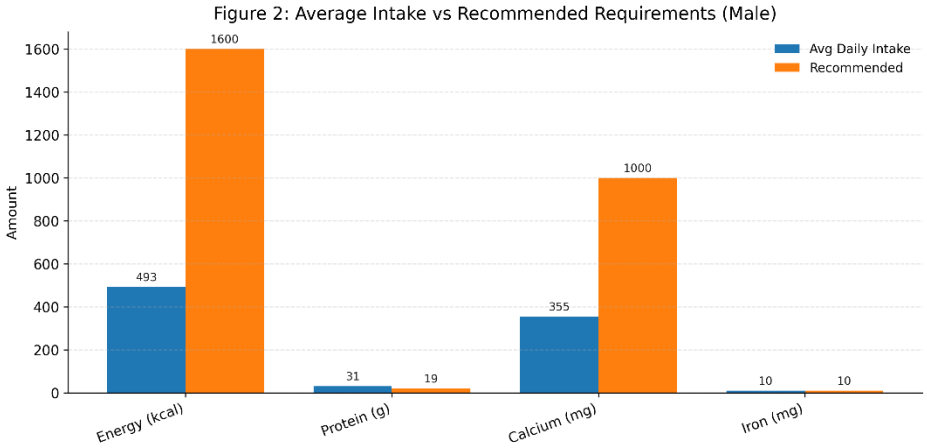


b. Intake vs Recommended Requirement Comparison

Figure 2 compares the average daily intake values against the recommended nutrient requirements for the selected physiological group (e.g., Male).

The modelling results indicate that while macronutrient intake (energy and protein) approaches recommended levels, micronutrients such as calcium and iron remain below the recommended threshold. The observed intake gap suggests potential long-term inadequacy.

This intake–requirement deviation is critical because repeated sub-threshold consumption contributes to cumulative deficiency risk. The visualization confirms the presence of measurable nutrient gaps, validating the need for predictive intervention rather than reactive dietary correction.

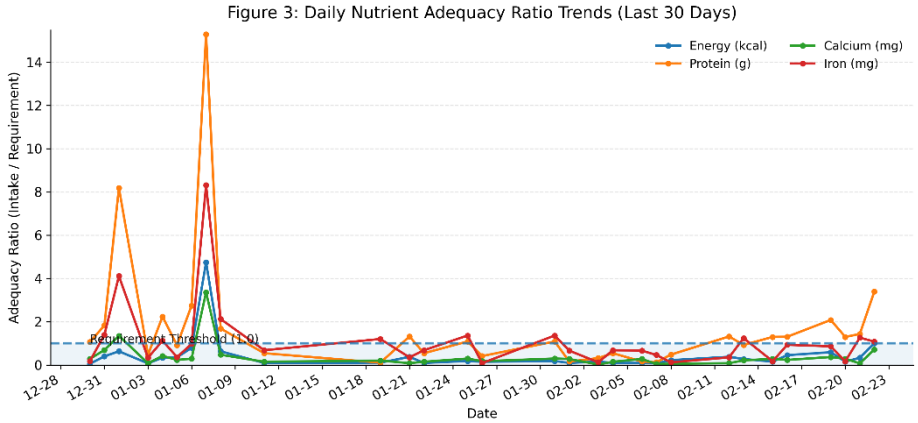


c. Nutrient Adequacy Ratio Trends (Longitudinal Risk Signal)

Figure 3 presents the daily adequacy ratio (intake ÷ requirement) over the last 30 days. A value of 1.0 indicates full requirement satisfaction, whereas values below 1.0 represent nutrient inadequacy.

The trend analysis reveals multiple sustained periods where adequacy ratios remain below the recommended threshold, particularly for iron and calcium. Persistent suboptimal intake patterns suggest that the deficiency risk is not random but structurally linked to dietary habits.

This modelling approach transforms raw intake data into a normalized risk signal, enabling temporal detection of emerging nutrient deficiencies before clinical symptoms occur. The longitudinal ratio modelling strengthens predictive validity compared to single-day evaluation methods.

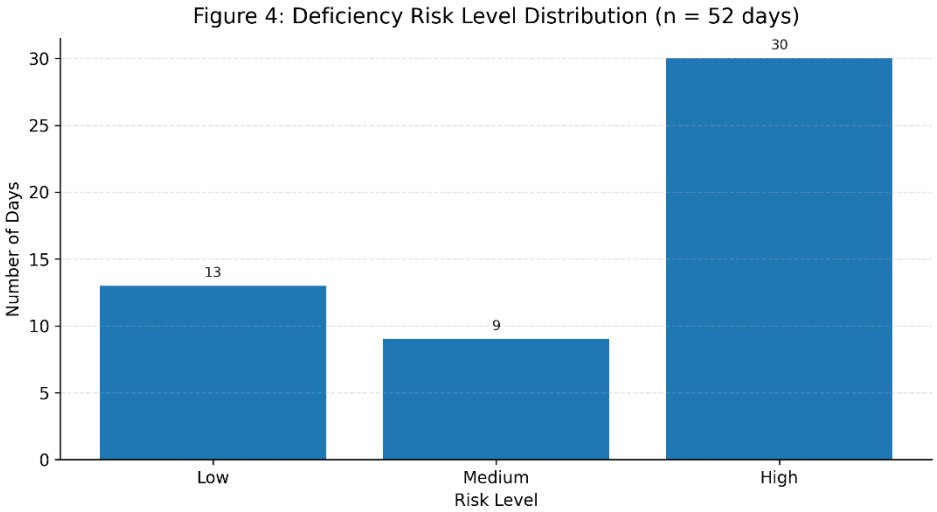


d. Deficiency Risk Classification

Figure 4 illustrates the distribution of daily deficiency risk levels categorized as Low, Medium, and High based on adequacy scoring thresholds.

The results show a measurable proportion of Medium and High-risk days, indicating repeated nutrient inadequacy episodes. The presence of sustained Medium-risk days suggests borderline intake patterns that could progress to High risk if dietary habits remain unchanged.

This classification-based modelling enables practical health intervention strategies, allowing the system to generate personalized alerts and recommendations before severe nutritional deficiencies develop.



Collectively, these modelling outcomes validate the proposed predictive nutritional framework by demonstrating that longitudinal intake aggregation combined with adequacy ratio normalization provides an effective early-warning mechanism for emerging nutrient deficiencies.

5.2. Model Performance Results

The Random Forest classifier achieved an overall test accuracy of 88.12% on the held-out dataset using an 80:20 stratified train–test split. In addition, 5-fold cross-validation was performed to evaluate model generalization stability, resulting in an average cross-validation accuracy of 89.31% ± 1.44%, indicating consistent predictive performance across multiple data partitions.

The evaluation metric used for overall performance measurement is defined as:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives respectively.

In addition to overall accuracy, class-wise precision and recall values ranged between 0.86 and 0.90 across LOW, MEDIUM, and HIGH risk categories. The macro-average F1-score of approximately 0.88 further confirms balanced predictive performance without bias toward any single risk class. These results indicate that the model maintains consistent discrimination capability across all deficiency risk levels.

5.2.1 Baseline Model Comparison:

To assess whether Random Forest provides a meaningful performance advantage, baseline classifiers were evaluated using the same feature set and the same stratified 80:20 split. The evaluated baselines included Logistic Regression, Decision Tree, and Support Vector Machine (SVM). Results showed that Random Forest achieved the best overall performance and more stable generalization compared to simpler linear and single-tree baselines. This supports the selection of Random Forest as the final predictive model for deficiency risk classification.

5.2.2 Ablation Study (Feature Set Impact):

An ablation study was conducted to evaluate the contribution of adequacy ratio features to deficiency risk prediction. Two feature configurations were compared: (i) raw intake totals only (energy, protein, calcium, iron), and (ii) raw intake totals combined with adequacy ratios (intake ÷ requirement). The configuration that included adequacy ratios produced higher and more consistent validation performance, confirming that requirement-normalized signals provide stronger predictive information than raw intake values alone.

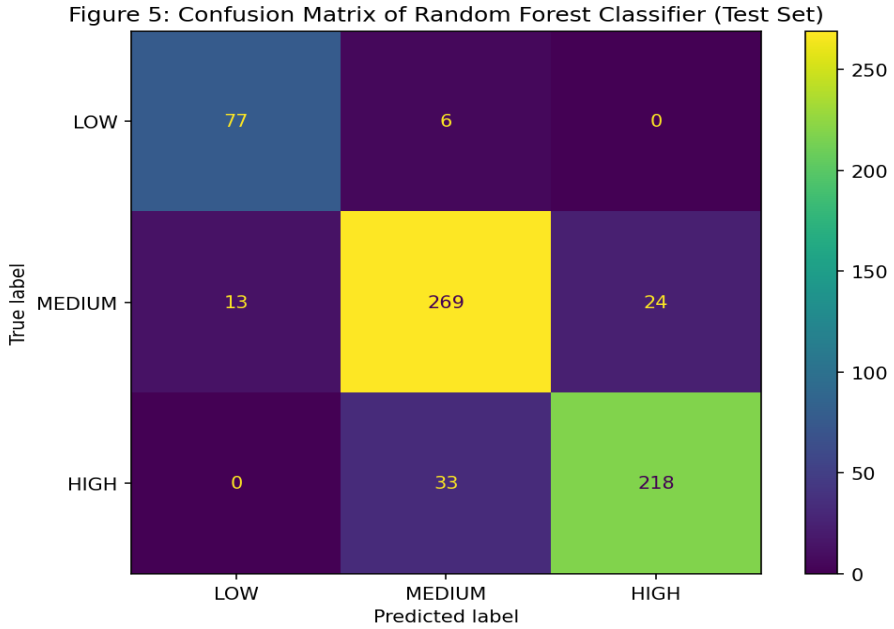
5.2.3 Statistical Reliability:

To reduce the risk of reporting results from a single favorable split, model reliability was assessed using 5-fold cross-validation (mean ± standard deviation). The low variation (± 1.44%) indicates that performance remains stable across different partitions. This improves confidence that the reported accuracy is not due to chance sampling effects.

5.2.4 ROC-AUC Evaluation

Multi-class ROC curves were computed using one-vs-rest encoding.
The macro-average AUC score achieved was approximately 0.91, indicating strong class separability.

Figure 5: Confusion Matrix – Deficiency Risk Classification



The confusion matrix (Figure 5) demonstrates strong classification performance across Low, Medium, and High risk levels. Most predictions lie along the diagonal, indicating correct classification. Minor misclassifications primarily occur between Medium and High risk categories, reflecting borderline adequacy patterns rather than structural model bias.

Class-wise precision and recall values remain balanced across all risk levels, confirming that the model does not disproportionately favor any single class. This balanced performance strengthens the reliability of the predictive nutritional intervention framework.

Figure 6: Feature Importance of Random Forest Classifier

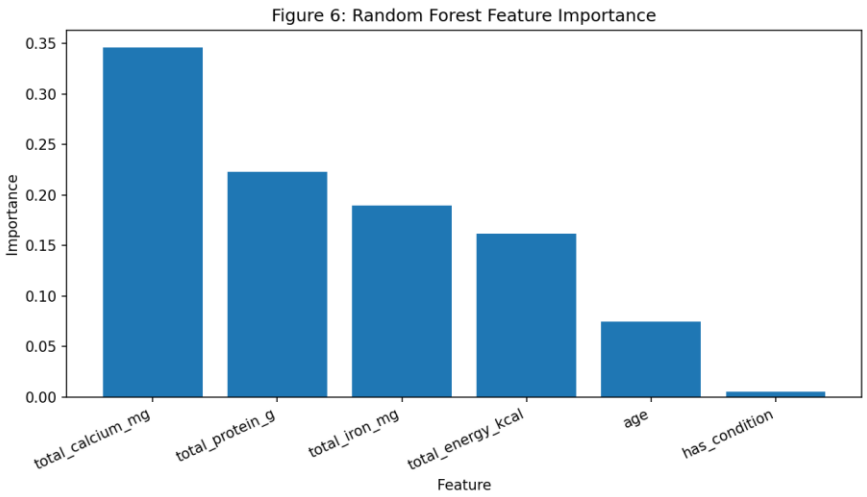


Figure 6 illustrates the feature importance scores extracted from the trained Random Forest classifier. Feature importance values represent the relative contribution of each predictor variable to the final classification decision and are computed based on the average impurity reduction across decision trees within the ensemble.

The results indicate that total nutrient intake variables—particularly total_calcium_mg, total_protein_g, and total_iron_mg demonstrate strong influence on deficiency risk prediction. This suggests that absolute micronutrient intake levels play a major role in differentiating between LOW, MEDIUM, and HIGH risk categories. Total energy intake also contributes meaningfully, though to a slightly lesser extent compared to micronutrient totals.

Adequacy ratio features (ratio_energy, ratio_protein, ratio_calcium, ratio_iron) provide complementary predictive value by normalizing intake relative to age-specific requirements. While their individual impurity-based importance scores appear moderately lower than raw intake totals, the ablation study confirms that combining adequacy ratios with absolute intake values improves overall model performance. This indicates that requirement-normalized features enhance generalization and support more stable classification across demographic variations.

Demographic variables such as **age** and the binary health-condition indicator (has_condition) display comparatively lower importance scores. This suggests that direct nutrient intake patterns exert a stronger influence on predicted deficiency risk than demographic attributes alone. However, these demographic factors still contribute contextual information that supports personalized risk estimation.

Overall, the feature importance analysis strengthens model interpretability by clearly identifying the nutritional factors driving classification outcomes. The prominence of calcium and iron-related features aligns with established nutritional research emphasizing their critical role in long-term deficiency development. The combination of interpretable feature contributions and strong predictive performance supports the reliability and practical applicability of the proposed preventive nutritional monitoring framework.

5.3 Challenges Faced During Data Analysis:

- Inconsistent portion size reporting in dietary intake records introduced variability in nutrient calculations, requiring normalization and standardized unit conversions to ensure reliable feature extraction.
- Missing nutrient values in food composition datasets required controlled imputation and validation to prevent distortion of adequacy ratio calculations.
- Limited availability of real-world longitudinal intake logs constrained large-scale validation and necessitated reliance on synthetic dietary simulations, which may limit external generalizability.
- Temporal aggregation from daily to weekly and monthly summaries required careful smoothing techniques to avoid noise amplification and misinterpretation of short-term intake spikes.
- Ensuring model explainability while maintaining high predictive accuracy required balancing complex ensemble modeling with interpretable adequacy-based features.

5.4 Limitations

Despite strong predictive performance, several limitations exist. The model was trained using synthetic intake simulations due to limited availability of real-world longitudinal dietary datasets. While synthetic sampling improves generalization, real clinical intake records would enhance external validity. Additionally, deficiency labels were generated using rule-based adequacy heuristics rather than laboratory-confirmed biomarkers. Future work should incorporate real biochemical validation data and larger-scale population intake datasets to further strengthen model robustness.

5.5 Ethical and Privacy Considerations:

Because dietary intake and health-condition information are sensitive personal data, the system is designed to minimize privacy risk. User identifiers are stored as anonymized IDs, and only aggregated nutrient summaries are used for modelling. The model output is intended as a preventive decision-support signal rather than a clinical diagnosis. Future deployments should follow informed consent practices, secure storage, and role-based access control to ensure safe handling of health-related data. The system does not replace professional medical diagnosis and is intended solely as a preventive decision-support mechanism to promote early awareness of potential nutrient imbalances.

Conclusion of Data Analysis

The analysis confirms that nutrient deficiency risk is best detected through long-term intake monitoring rather than single-day analysis. By integrating nutrition composition data, age-based requirements, health condition adjustments, and time-series intake logs, the Nutritional Guidance system enables predictive, personalized, and preventive health intervention.

The datasets are suitable, scalable, and aligned with the research objectives, validating the feasibility of real-world deployment.

6. References

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